

UCC502: TRANSPORTATION ENGINEERING

Prerequisite(s): NIL

L	T	P	Cr
3	0	2	4.0

Course Objective: The objective of the course is to enable students' study different aspects of transportation engineering such as to analyze & design various road & airport pavement structures using various software's, to differentiate between the various types of pavement layers & to perform the detailed geometric design for road, railways & airport projects using various computing tools.

Geometric Design: Introduction, Highways Classification, right of way, Width of formation, Sight Distances, stopping site distance, overtaking sight distance, overtaking zones, camber, Road Curves, Transition Curves, Super elevation, Widening at curves, IRC codal recommendations for various geometric design parameters, Application of Mx-Road Software for the geometric design, Geometric design for railways & airport projects, Road Safety Audits, Highway capacity & Intersection design. Road user characteristics, Importance of traffic studies, spot speed, speed and delay and origin and destination studies.

Design & Construction: Factors affecting design of pavements, design principles & design procedures as per IRC 37 guidelines, Design of PQC pavements as per IRC 58 guidelines, Use of software's IITPAVE, KENPAVE for design of highway pavements, Design of airport pavements as per FAA guidelines, Application of FAARFIELD software for design of airport pavements. Specifications & construction method statements for various pavement layers & permanent way section. Importance & Principles of Drainage, Surface Drainage & Sub-Surface drainage

Failures of flexible and rigid pavements & Highway Maintenance: Causes of Failures and Remedial Measures, Maintenance of flexible and rigid pavements, pavement evaluation and its strengthening method, overlay design using IRC 81 & IRC115 guidelines, use of software's kgpback & ELMOD for overlay design of highway pavements, LCN-PCN concept for airport pavements, COMFAA software for airport pavements evaluation, concept of thin & ultrathin white toppings, design & scheduling of maintenances activities, development of Pavement maintenance management systems for different categories of roads, Economic Evaluation of Highway Projects and Life Cycle Costing Concept using HDM-4 software / any other soft computing techniques.

Laboratory Work: The students will perform various quality control tests as per Indian Road Congress (IRC) & Ministry of Road Transport & Highways (MORTH) specifications for the various layers of the pavement section. Paving bitumen & bituminous mix testing like penetration value, softening point, viscosity & binder rheology, ductility value, centrifugal extraction, aggregate strength tests, pavement layer gradation & mix designs for various pavement layers, Deflection studies for pavements.

Software's application for the course:

- i IITPAVE
- ii KENPAVE
- iii KGPBACK
- iv ELMOD
- v FAARFIELD
- vi COMFAA
- vii Mx ROAD

Experimental Project/assignment/Micro Project: Students in groups of 4 to 6 will do the projects:

1. Design the flexible pavement as per IRC 37 guidelines for the given traffic data using various software's.
2. Design the rigid pavements as per the IRC 58 guidelines for the given traffic data using various software's.
3. Pavement Maintenance Management system using HDM-4 / other soft computing techniques.

Course learning Outcome (CLO):

Upon completion of this course, the students will be able to:

1. Quantify the specifications for various construction materials / pavement layers / permanent way layers.
2. Perform geometric design for highways, railways & airport projects using various software's.
3. Perform analysis and design of flexible and rigid pavements for highway & airport pavements using various software's.
4. Develop the pavement maintenance management systems & perform the life cycle cost analysis of the highway / airport projects using HDM-4 / other soft computing techniques.

Text Books:

1. Khanna S.K. and C.E.G. Justo, *Highway Engineerin*, Nemchand Bros (2017).
2. Kadyali L. R.; *Highway Engineering*, Nem Chand & Brothers, Roorkee (2018).

Reference Books:

1. Sharma & Sharma; *Principle and Practice of Highway Engineering*, Asia Publishing House, New Delhi (2010).
2. G.V.Rao, Tata McGraw Hill, New Delhi "Principles of Transportation and Highway Engineering" (1996)
3. Yoder E. J.; *Principles of Pavement Design*, John Wiley & Sons (2011)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30